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EDUCATION

- **Universitat Politècnica de Catalunya** Barcelona, Spain
Master of Numerical Methods in Engineering Sept 2022 - May 2025
Courses: Advanced Fluid Mechanics, HPC, Machine Learning, Artificial Intelligence, Structural Dynamics, Parametric Model Order Reduction (PMOR)
- **JSS Academy of Technical Education** Bangalore, India
Bachelor of Engineering - Mechanical Engineering Aug 2015 - June 2019
Courses: Fluid Dynamics, FEA, Thermodynamics, Heat Transfer, Turbo Machinery

SKILLS SUMMARY

- **Modeling:** CATIA, Solidworks, Spaceclaim, HyperMesh, AutoCAD
- **CAE Tools:** ANSYS, Star-CCM+, Abaqus, OpenFoam, Fluent, Workbench
- **Programming Languages:** Python, C++, MATLAB, TCL/TK
- **Languages:** **Native/ Bilingual Proficiency** - English, **Limited Proficiency** - Spanish, German
- **Soft Skills:** Leadership, Task Management, Writing, Public Speaking, Time Management

WORK EXPERIENCE

- **Limak Construction Europe S.L.** Dec 2025 - Present
Project Control Engineer
 - **MEP Mechanical Technical Engineering:** Supporting HVAC, plumbing, and fire protection scopes through technical review, coordination, and compliance within the MEP department.
 - **Excel-Driven Technical & Cost Control:** Managing advanced Excel trackers for quantities, costs, progress, and contractor data to support accurate reporting and control.
- **Mekano4** March 2025 - May 2025
Structural Simulation Engineer - Intern
 - **Bridge Component Simulation:** Executed nonlinear structural simulations using ANSYS to assess the behavior of post-tensioned systems, incorporating geometric and material nonlinearity for critical bridge elements.
 - **Contact & Convergence Modeling:** Developed and refined models with complex contact interactions (wedge-anchor-cable interfaces), implementing advanced solver strategies to ensure accurate convergence and displacement control.
- **CSolar.es** Oct 2024 - Dec 2024
Simulation Engineer - Intern
 - **Solar Panel CFD Analysis:** Conducted comprehensive CFD simulations to analyze airflow performance over solar panels, optimizing efficiency and durability.
 - **Structural Simulation Expertise:** Performed advanced structural analyses using Abaqus to evaluate and enhance the structural integrity of solar panel systems.
 - **Industry Presentation:** Presented key findings and technical insights at an ANSYS event, demonstrating expertise in simulation methodologies and their real-world applications.
- **Faurecia Clean Mobility** June 2022 - Sep 2022
Senior CFD Engineer - Full Time
 - **Exhaust System Optimization:** Led team to improve exhaust system simulation efficiency using CAE tools and advanced programming.
 - **Parametric & Comparative Analysis:** Performed parametric studies and comparative analysis to enhance simulation accuracy and insights.
 - **Multidisciplinary Collaboration:** Scoped projects with field engineers, delivering professional services while ensuring high-quality client support.
- **Copes Tech India Pvt. Ltd** Jan 2020 - May 2022
CFD Analyst - Full Time
 - **CFD & Thermal Analysis:** Conducted fluid flow and thermal CFD analysis of engine exhaust systems, with Star-CCM+ and Hypermesh.
 - **Simulation Optimization:** Led geometry optimization, mesh modeling, and structural dynamics simulations.
 - **Advanced Simulation Tools:** Utilized Star-CCM+, Hypermesh, and Ansys to develop comprehensive solutions, optimize designs, and enhance functionality and efficiency.
- **Copes Tech India Pvt. Ltd** Jul 2019 - Jan 2020
CFD Engineer - Full Time
 - **Advanced CFD Simulation:** Executed 30+ 1D, 2D, and 3D simulations, showcasing expertise in fluid flow and thermal analysis with high accuracy and efficiency.
 - **Technical Proficiency:** Demonstrated advanced skills in modeling, meshing, and simulation optimization using tools like Star-CCM+, Hypermesh, and Ansys.

- BHEL-EDN** Nov 2018 - Dec 2018
 • *Simulation Engineer - Intern*
 - **Cold Plate Design & Testing:** Designed and rigorously tested a cold plate for traction applications, showcasing innovation and practical problem-solving capabilities.
 - **Advanced Modeling:** Created a detailed SolidWorks model to support comprehensive analysis and enhance traction system performance.
 - **Thermal Management Expertise:** Applied engineering principles to optimize cold plate design, ensuring efficient heat dissipation and reliability in traction systems.
- Triveni Engineering & Industries Limited** Jul 2018 - Aug 2018
 • *Intern*
 - **Gear Manufacturing Expertise:** Gained hands-on experience in the complete gear manufacturing process, including raw material selection, production, and testing, with a focus on practical engineering applications.

Projects

- Development of a Machine Learning Tool for Fake News Detection (Oct 2023 - Dec 2023)** *Tools: Python, Scikit-learn, Logistic Regression, Random Forest, Gradient Boosting*
 - **Implemented TF-IDF for Feature Extraction** - Preprocessed and transformed raw text data into numerical form using TF-IDF to capture word importance across documents.
 - **Built Machine Learning Models** - Trained models like Logistic Regression and Random Forest to classify news articles, selecting Random Forest for its superior accuracy.
 - **Improved Model Robustness and Accuracy** - Conducted hyperparameter tuning and cross-validation, achieving 85% accuracy and reducing model overfitting on test data.
 - **Deployed Solution for Real-World Use** - Deployed the trained model to classify news articles in real time, providing a robust and scalable fake news detection tool.
- Reduced Order Modeling for Nonlinear Structural Dynamics (Jan 2025 – May 2025)** *Tools: Python, NumPy, SciPy, SVD, Operator Inference, Newmark Method, Kratos Multiphysics*
 - **Developed Non-Intrusive Reduced Order Models (ROM)** – Implemented a data-driven ROM using Operator Inference and Proper Orthogonal Decomposition (POD) to reduce computational cost in structural dynamics simulations.
 - **Simulated Cantilever Beam Dynamics** – Performed full-order simulations in KratosMultiphysics and solved reduced models using Newmark- β with Newton-Raphson iteration for nonlinear systems.
 - **Validated ROM Accuracy** – Evaluated model performance by comparing displacement, velocity, and acceleration results with high-fidelity full-order models, achieving over 98% dimensionality reduction.
- Simulation of Turbulent Flow Through a Pipe (Jan 2022 - Mar 2022)** *Tools: MATLAB, OpenFOAM, Python, k-epsilon/k-omega Models*
 - **Conducted CFD Simulations in OpenFOAM** - Simulated turbulent flow in a circular pipe using Reynolds-Averaged Navier-Stokes (RANS) equations to predict flow behavior.
 - **Automated Mesh Generation** - Created efficient meshes with automated tools, improving simulation accuracy and computational efficiency.
 - **Validated Turbulence Models** - Compared k-epsilon and k-omega models against theoretical velocity profiles to evaluate their performance for internal flow predictions.
- Simulation of Supersonic Flow in Nozzles (Sep 2021 - Nov 2021)** *Tools: MATLAB, MacCormack Method, Compressible Flow Analysis*
 - **Developed Solver for Compressible Flow** - Implemented the MacCormack method to solve compressible Navier-Stokes equations for supersonic flow in convergent-divergent nozzles.
 - **Analyzed Shock Wave Formation** - Simulated shock waves and expansion fans occurring within the nozzle due to pressure and Mach number variations.
 - **Validated Numerical Results** - Compared simulation results with isentropic flow theory, ensuring solver accuracy and highlighting flow behavior under supersonic conditions.

PUBLICATIONS

- **A Literature Study on Automobile Fuel/Propulsion System Technologies:** Published on: 31 July 2017. ISSN: 2320-5547
- **A Literature Survey on Automobile fuel system technologies:** Published on: 31 July 2017. ISSN: 2320-5547
- **National Conference on Emerging Trends in Material, Design, and Manufacturing (NCMDM- 2019):** Characterization of E Glass Fiber reinforced with Aluminum 8176 metal matrix composite, B.E. Project